

Tide retrospective:
perturbed harmonic-Gibbs moments $\langle x^k \rangle_h$ for $k = 3, 4$ (O1)

Daniel Murfet

7 May 2026

Abstract

This retrospective records a 528-line programme continuation of I4 (global-constancy of the harmonic Gibbs covariance, merged earlier today). I4 closed $k = 1, 2$: $\langle x \rangle_h = -h/\lambda$ and $\langle x^2 \rangle_h = 1/(\lambda t) + h^2/\lambda^2$. This tide extends to $k = 3, 4$ via the same square-completion shift method, and lands the ****public Gibbs-expectation transport lemma**** GPT-5.5 Pro named at the deliberation: $\langle f(x) \rangle_h = \langle f(y - h/\lambda) \rangle_0$ for any observable f . The decisive mathematical observation: the normalised perturbed law is exactly $\mathcal{N}(-h/\lambda, 1/(\lambda t))$, so the closed forms are the raw Gaussian moments. Single-session run, build green after three small fixes (a cast normalisation for $((2 \cdot 2 - 1)!!)$ in $\int y^4$, an rpow-algebra sequencing in $\int y^4$, and the standard `Pi.add-vs-single-lambda` workaround per the laplace `CLAUDE.md`). Zero `sorry`, zero `axiom`, zero `native_decide`.

1 Setting

I4 (global-constancy harmonic, merged this morning) proved that the harmonic Gibbs covariance is globally constant in h : $\text{Cov}_h(x, x) = 1/(\lambda t)$ for all $h \in \mathbb{R}$. Its proof routed through closed-form evaluations of three perturbed integrals via a square-completion shift identity. I4's retrospective §Follow-ups 5 explicitly flagged: *“Promote `integral_with_perturbation_eq_shifted` on second use.”*

This O1 tide is that second use. The user's direction was the family $\langle x^k \rangle_h$ for $k \in \{1, 2, 3, 4\}$, with explicit closed forms named by binomial expansion of $(y - h/\lambda)^k$. The deliberation converged on Candidate A+ (specific theorems for $k = 3, 4$ *plus* a public shift-at-expectation transport lemma) rather than B (a parametric general- k statement) or C (just $k = 3$). Both Claude and GPT-5.5 Pro voted A+ on the first round.

2 The thing formalised

For the harmonic Gibbs measure $\mu_{t,h}(x) \propto \exp(-t((\lambda/2)x^2 + h \cdot x))$ on $\mathbb{R}$ against Lebesgue with $\lambda, t > 0$, the perturbed moments are

$$\langle x^3 \rangle_h = -\frac{3h}{\lambda^2 t} - \frac{h^3}{\lambda^3}, \quad \langle x^4 \rangle_h = \frac{3}{\lambda^2 t^2} + \frac{6h^2}{\lambda^3 t} + \frac{h^4}{\lambda^4}.$$

The Lean signatures (public):

```

theorem gibbsExp_harmonic_h_eq_unperturbed_shift
  {lam t : ℝ} (hlam : 0 < lam) (ht : 0 < t)
  (h : ℝ) (f : ℝ → ℝ) :
  Threepoint.gibbsExp (volume : Measure ℝ)
    (fun x lam / 2 * x ^ 2)
    (fun x x) t h f
  = Threepoint.gibbsExp (volume : Measure ℝ)
    (fun x lam / 2 * x ^ 2)
    (fun x x) t 0
    (fun y f (y - h / lam))

theorem gibbsExp_h_cube_harmonic_eq
  {lam t : ℝ} (hlam : 0 < lam) (ht : 0 < t) (h : ℝ) :
  Threepoint.gibbsExp ... t h (fun x x ^ 3)
  = -3 * h / (lam ^ 2 * t) - h ^ 3 / lam ^ 3

theorem gibbsExp_h_quartic_harmonic_eq
  {lam t : ℝ} (hlam : 0 < lam) (ht : 0 < t) (h : ℝ) :
  Threepoint.gibbsExp ... t h (fun x x ^ 4)
  = 3 / (lam ^ 2 * t ^ 2) + 6 * h ^ 2 / (lam ^ 3 * t)
    + h ^ 4 / lam ^ 4

```

3 Proof strategy

The decisive observation, due to GPT-5.5 Pro: the normalised perturbed law is exactly Gaussian with mean $-h/\lambda$ and variance $1/(\lambda t)$. So the moments are the standard raw Gaussian moments $E[X^k] = \mu^k + \binom{k}{2}\mu^{k-2}\sigma^2 + \binom{k}{4}\mu^{k-4}\sigma^4 + \dots$ (only even-power-of- σ terms survive by parity). Substitute $\mu = -h/\lambda$, $\sigma^2 = 1/(\lambda t)$ and the closed forms drop out.

The Lean proof assembles in three layers.

Layer 1: re-derive the I4 shift identity locally. The integral-level shift identity is **private** in the I4 file. To keep this tide self-contained, I re-prove it under a primed name. The proof is verbatim from I4: square-completion gives the pointwise identity $\exp(-t(L_h(x))) = \exp(th^2/(2\lambda)) \cdot \exp(-t(L_0(x + h/\lambda)))$, which integrates to $\int f(x)e^{-tL_h}dx = e^{th^2/(2\lambda)} \int f(y - h/\lambda)e^{-tL_0}dy$ via `integral_add_right_eq_self` for the shift. About 30 lines.

Layer 2: integrability witnesses + closed-form moments. Build a single $\int y^n e^{-(\lambda/2)y^2}$ integrability lemma for any n via Mathlib’s polynomial-times-Gaussian integrability result, after a coefficient-matching rewrite.¹ Build closed forms for $n = 0, 1, 2, 3, 4$ by specialising Tide 10’s even/odd moments.² n odd gives 0; $n = 0$ gives $\sqrt{2\pi/(\lambda t)}$; $n = 2$ gives $(1/(\lambda t))\sqrt{2\pi/(\lambda t)}$ via $(\lambda t)^{-(1+1/2)} = (\lambda t)^{-1} \cdot (\lambda t)^{-1/2}$; $n = 4$ gives $(3/(\lambda t)^2)\sqrt{2\pi/(\lambda t)}$ via the same rpow-additivity argument with $(\lambda t)^{-(2+1/2)}$. About 100 lines for the five bridges.

Layer 3: the transport lemma and the two moments. The transport lemma’s proof is short: apply the shift identity to both numerator and denominator of $\langle f \rangle_h$. Each carries an $\exp(th^2/(2\lambda))$

¹Mathlib lemma: `integrable_pow_mul_exp_neg_mul_sq`.

²`harmonic_int_pow_even` and `harmonic_int_pow_odd`.

factor; `field_simp` cancels them in the ratio, delivering the unperturbed expectation of the shifted observable. The cube and quartic theorems then expand $(y - h/\lambda)^k$ pointwise via `funext y; ring`, split the integral via `integral_add` and `integral_const_mul`, plug in the $n = 0, 1, 2, 3, 4$ closed forms, and finish with `field_simp` (no trailing `ring` needed — `field_simp` closes the rational-function identity outright). About 90 lines per moment.

4 Roadblocks and resolutions

Three small fixes between draft and green build, none load-bearing.

Cast normalisation in $\int y^4$. The `harmonic_int_pow_even` statement carries $((2k - 1)!) : \mathbb{R}$ as a coefficient. At $k = 2$ this is $((2 \cdot 2 - 1)!) = 3! = 3$, but the cast needed explicit unfolding (a `change` to $((3 : \mathbb{N})!) : \mathbb{R}$), then `decide` on the natural-number `doubleFactorial`, then `norm_num` on the cast). The $((2 : \mathbb{N}) : \mathbb{R}) = 2$ exponent normalisation likewise needed an explicit `norm_cast`. About 5 lines of cast plumbing per even-power closed form.

rpow algebra in $\int y^4$. The closed form $(\lambda t)^{-(2+1/2)} = (\lambda t)^{-2} \cdot (\lambda t)^{-1/2}$ — straightforward mathematically — is fragile in Lean because the parenthesisation of `rpow_add` matters and the goal can have several subterms matching the rewrite pattern. First attempt with a chained rewrite sequence failed at *“Tactic rewrite failed: did not find an occurrence of the pattern”* on the third step. Fix: factor out a single named `hcombine` hypothesis with `Real.rpow_add` inside, then `rw [hcombine]; ring` closes the goal in one shot. The lesson: when chaining `rpow` rewrites, the named-intermediate form is more robust than the inline-show pattern.

Pi.add vs single-lambda mismatch in `integral_add`. Standard issue per the `laplace CLAUDE.md`: `Integrable.add` returns `Integrable (f + g)` where the sum is `Pi.add`, but `integral_add`’s rewrite pattern is the single-lambda form `fun y => f y + g y`. Lean’s pattern-matcher doesn’t unfold `Pi.add` automatically, so the rewrite fails on the elaborated goal. Fix: introduce a type-ascribed single-lambda integrability witness `have h_combined : Integrable (fun y => ...)` `volume := h_left.add h_right` before the `rw [integral_add ...]`. This issue surfaced in both the cube and quartic numerator integrals; the fix added six `have` blocks across the two theorems.

Trailing ring after `field_simp`. Both the cube and quartic theorems’ final lines originally ended with `field_simp; ring`. After fixing the `Pi.add` issue, the `field_simp` step closed the goal completely, leaving no goals for the trailing `ring`. Removing the dead `ring` fixed two *“No goals to be solved”* errors.

5 What was Mathlib, what was new

Mathlib. The Bochner-integral linearity lemmas (translation invariance, additivity, scalar multiplication, congruence under a.e. equality), the polynomial-times-Gaussian integrability result, and the standard `rpow` algebra (additivity in the exponent, negation, `sqrt-as-rpow`, `Nat-cast`).³ Plus `field_simp` and `ring` as closing tactics. All familiar; no new Mathlib API surfaced.

³Names: `integral.add_right_eq_self`, `integral.const_mul`, `integral.add`, `integral.sub`, `integral.congr_ae`, `integrable.pow_mul_exp_neg_mul_sq`, `Real.rpow_add`, `Real.rpow_neg`, `Real.sqrt_eq_rpow`, `Real.rpow_natCast`.

Laplace seabed (existing). I4’s integral-level shift identity (re-derived locally, since the I4 original is `private`). Tide 10’s even/odd unperturbed-moment closed forms for $n = 0, 1, 2, 3, 4$ and `partitionFunction_harmonic`. The standard `Threepoint.gibbsExp` definition.

New. Five $\int y^n e^{-t(\lambda/2)y^2}$ closed-form bridges (named `int_pow_zero/one/two/three/four`), one general polynomial-times-Gaussian integrability witness parameterised by $n : \mathbb{N}$, the public shift-at-expectation transport lemma, and the two public moment theorems. The transport lemma is the load-bearing new public API: it captures the perturbed-law-equals-shifted-unperturbed-law fact in a form independent of the choice of observable, and is the natural lift of I4’s integral-level shift identity to the expectation level.

6 Lessons

- *The transport lemma was the right abstraction.* GPT-5.5 Pro named it at the deliberation, replacing my draft’s plan to promote the integral-level shift identity. The transport lemma’s stronger invariant — *the perturbed law is the shifted unperturbed law, full stop* — yields shorter cube/quartic proofs (no need to manage the $\exp(th^2/(2\lambda))$ factor explicitly at every site; `field_simp` cancels it once inside the transport lemma’s proof).
- *Re-deriving I4’s private helpers is cheaper than un-privating.* The integral-level shift identity is ~ 30 lines. Modifying the I4 file to expose it would have crossed the “edit-merged-file” boundary and risked merge surprises with the L2 in-flight tide that branched off N1 (which itself was just merged). Local re-derivation kept the change contained.
- *The `Pi.add` workaround is a recurring pattern.* This is the third tide today (after I4 itself and G2+) where the `Integrable.add-returns-Pi.add` vs. `integral_add-expects-single-lambda` mismatch surfaced. The laplace `CLAUDE.md` already documents the workaround; future tides should reach for the type-ascribed-single-lambda witness on first sight, not after a failed rewrite.
- *One parametric integrability witness beats three specialised ones.* I4 used three separate witnesses for the unperturbed Gaussian moments (zeroth, first, second power). This tide uses one parametric witness taking $n : \mathbb{N}$, derived from Mathlib’s polynomial-times-Gaussian integrability via a coefficient rewrite. Worth promoting to lean-common when a third consumer materialises.

7 Follow-ups

- *Generic- k moments via the binomial sum.* Given the parametric integrability lemma above, a single `gibbsExp_h_pow_harmonic_eq` for any $k : \mathbb{N}$ (Candidate B from this tide’s deliberation, deferred) would land in ~ 80 additional lines: $\sum_{j=0}^k \binom{k}{j} (-h/\lambda)^{k-j} \langle y^j \rangle_0$, with a separate `gibbsExpectation_harmonic_pow_*_y` lemma for general $\langle y^j \rangle_0$. Defer until a $k = 5$ or $k = 6$ consumer arrives.
- *Translation invariance for arbitrary potentials.* The shift identity here is harmonic-specific (square completion). A generic potential L admits a transport lemma of the form $\langle f(x) \rangle_{L,h} = \langle f(y - \tau(h)) \rangle_{L,0}$ only when L has appropriate symmetry. Worth recording as a *not-tide-shaped* candidate: depending on the symmetry hypothesis bundle, this could be a 200-line expansion or a multi-tide programme.

- *\leanref-tag the primer*. The primer’s discussion of the shifted-Gaussian centroid mentions $\langle x^3 \rangle_h$ and $\langle x^4 \rangle_h$ as exercises in perturbative expansion. Pin `\laplaceCommit = ea3e13e` (this tide’s commit) and tag both. Pairs with K2 / N4 in the survey.
- *Discharge mixed-power GibbsObservable hypotheses (O2 in the survey)*. The transport lemma and the cube/quartic moments may discharge some of the mixed-power `GibbsObservable` hypotheses Tide 13 left conditional. Worth a follow-up audit.

Acknowledgements

GPT-5.5 Pro’s deliberation was decisive on three counts. First, the recasting of the closed-form derivation in terms of Gaussian raw moments $E[X^k] = \mu^k + \binom{k}{2}\mu^{k-2}\sigma^2 + \dots$ under $X \sim \mathcal{N}(-h/\lambda, 1/(\lambda t))$ replaced the binomial-of- $(y - h/\lambda)^k$ derivation in my draft with the shorter Gaussian-moment derivation. Second, the recommendation to land a public Gibbs-expectation transport lemma rather than just the integral-level shift identity. Third, a numerical-sanity table of five (λ, h, t) cases and their closed-form values for verification (which matched `scipy.quad` to machine precision). Full consult preserved at the GPT v1 file.⁴

This Tide branched off the post-N1-merge `main` (commit `47e3426`). I4 is at `5aa212c`; the transport lemma re-derives I4’s private shift identity locally rather than modifying the I4 file. No in-flight tides interact with this work (L2 / Q1 / A2 all on parallel sessions, all independent of the perturbed harmonic moments).

⁴Preserved at `projects/primer/tide-log/gpt55_tide_perturbed_moments_harmonic_v1.md`.